

STAR: Spliced Alignment

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Introduction

STAR (Spliced Transcripts Alignment to a Reference) is the de facto standard short-read RNA-seq aligner. It is designed specifically for the splice-aware alignment problem: reads from mature mRNA cross exon-exon junctions, and a naive aligner that requires contiguous matches misses these split alignments and misclassifies them as multi-mappers or unaligned. STAR's seed-extend algorithm splits reads at candidate junctions, aligns the segments to the reference genome, and assembles them into spliced alignments — including across novel (unannotated) junctions.

Prerequisites

A working understanding of RNA-seq experimental design, the difference between mRNA and DNA sequencing, and basic familiarity with sequence alignment concepts.

Theory

STAR's two-pass mapping is the standard mode: the first pass aligns each read independently and accumulates a list of candidate splice junctions; the second pass adds those junctions to the reference index and re-aligns, producing more accurate spliced alignments especially across novel junctions. The output is a BAM file plus a tab-separated splice-junction summary (`SJ.out.tab`).

STAR also supports chimeric alignment for fusion-gene detection (reads spanning two distinct genomic loci), expectation-maximisation-based multi-mapper resolution, and stranded alignment when library protocols preserve transcript orientation.

Assumptions

Short RNA-seq reads (typically 50–150 bp Illumina); a reference genome and (preferably) a gene annotation in GTF format; sufficient memory — STAR's index for human genome takes about 30 GB of RAM.

R Implementation

```
library(Rsubread)

# Rsubread::align is Subjunc (splice-aware) or align
# index <- "subread_index"
# buildindex(basename = index, reference = "genome.fa")

# Splice-aware alignment
# align(index = index, readfile1 = "reads.fastq.gz",
#       output_file = "aligned.bam",
#       type = "rna", useAnnotation = TRUE,
#       annot.inbuilt = "hg38")
```

Output & Results

The primary output is a BAM file containing spliced alignments and a `SJ.out.tab` file summarising splice junctions detected (including their support count and annotation status). For typical RNA-seq experiments, 90–95 % of reads align uniquely; a much lower rate signals contamination, library-prep problems, or annotation mismatch.

Interpretation

A reporting sentence: “STAR (v2.7.10a) two-pass alignment to GRCh38 with the GENCODE v44 annotation aligned 94 % of reads uniquely; 2.3 million splice junctions were detected, of which 98 % matched annotated junctions, suggesting limited novel splicing in the experimental conditions.” Always report the genome assembly, annotation version, and unique-mapping rate.

Practical Tips

- STAR needs about 30 GB of RAM for the human genome index; for smaller genomes (yeast, bacteria), it runs comfortably on a laptop.
- Two-pass mode (`twopassMode = "Basic"`) is standard for novel-junction detection; the first-pass-only mode is faster but less accurate.
- For transcript-level quantification, pair STAR alignments with RSEM, Salmon, or Kallisto; the BAM is the input to abundance estimation.
- HISAT2 is a faster, lower-memory alternative aligner with similar accuracy; use it when memory is constrained.
- Pseudo-aligners (Salmon, Kallisto) skip the BAM-producing alignment entirely; faster but less suitable when isoform-level alignments are needed (variant calling, fusion detection).
- Use a reference GTF annotation to improve junction accuracy; novel-junction detection works without it but is more error-prone.

R Packages Used

Rsubread provides splice-aware alignment via `subjunc` and feature counting via `featureCounts`; Rsamtools and GenomicAlignments for downstream BAM manipulation; tximport for importing transcript-level abundance estimates from Salmon, Kallisto, or RSEM into Bioconductor; STAR itself is a command-line tool but its output integrates with all standard R/Bioconductor RNA-seq workflows.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat